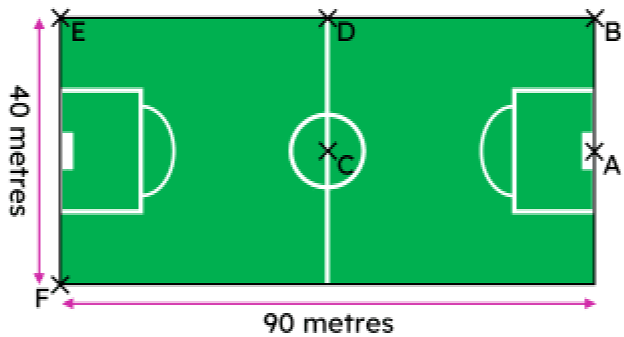
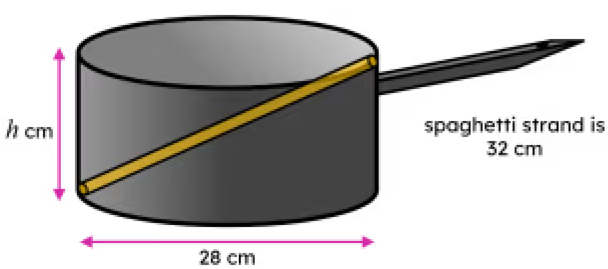
Pythagoras' theorem in context

1 On this football pitch, the goalkeeper (at point A) wants to kick the ball to a striker (at point E), however the goalkeeper's longest kick is 91 m. Which of these statements are correct? Tick **2** correct answers



- [ ] The distance between the goalkeeper and the striker is 49.24 metres.
- [x] The distance between the goalkeeper and the striker is 92.20 metres (to 2 d.p.).
- [ ] The distance between the goalkeeper and the striker is 98.49 metres (to 2 d.p.).
- [ ] The ball the goalkeeper kicks will reach the striker.
- [x] The ball the goalkeeper kicks will not reach the striker.

2 A factory produces cylindrical saucepans whose maximum base length is 28 cm. What must the minimum integer height of the saucepan be (in cm) so it can fully submerge a spaghetti noodle. Tick **1** correct answer



- [ ] 15 cm
- [x] 16 cm
- [ ] 19 cm
- [ ] 28 cm
- [ ] 32 cm

3 An archer fires an arrow that travels 20 metres to hit a target that is 9 m above ground. The arrow then drops to the ground. The archer must walk <u>17.86</u> m (to 2 d.p.) to pick up the arrow. Fill in the blank

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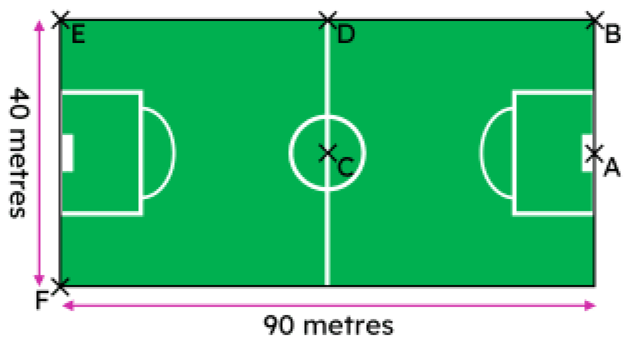
4 An archer fires an arrow at a target 20 m away. The arrow drops 9 m to the ground. Jun uses Pythagoras' theorem to find how far the archer walks to collect the arrow. What assumptions does Jun make? Tick **3** correct answers

- [x] The arrow travels in a straight line.
- [ ] The speed of the arrow is constant.
- [x] The ground the archer walks on to pick up the arrow is flat.
- [x] The archer fires the arrow from the ground.
- [ ] The accuracy of the archer.

5 The captain (at C) passes the ball to the goalkeeper (at A), who then passes to the midfielder (at D), who then passes to a striker (at F). The total distance the ball travelled is 154.45 m (to 2 d.p.). Fill in the blank



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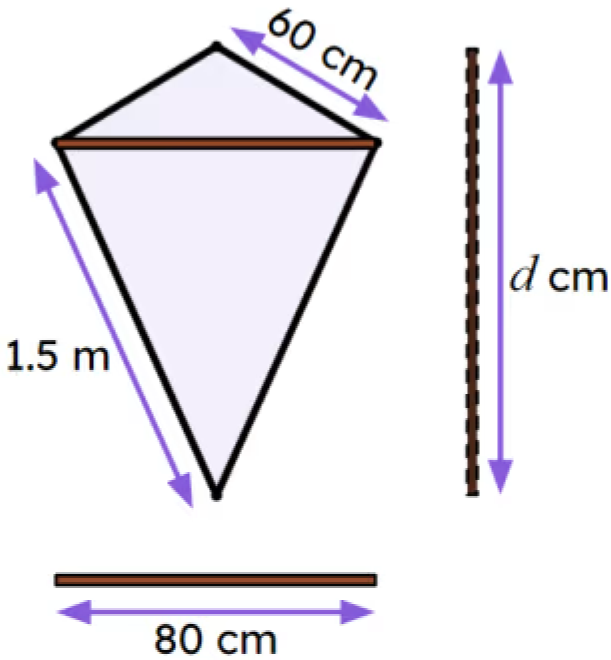
6 Two bamboo sticks are placed on the diagonal of this kite to support it during high winds. The length of the vertical bamboo stick is 189.29 cm (to 2 d.p.). Fill in the blank



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